

Acidic C-terminal domains autoregulate the RNA chaperone Hfq

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SITUATIONAL CONTEXT HOOK

In *Escherichia coli*, the RNA chaperone Hfq contains an intrinsically disordered C-terminal domain (CTD) that regulates RNA binding and duplex release. Using Rosetta de novo computational modeling and in vitro fluorescence anisotropy binding assays, the acidic tip of the *E. coli* Hfq CTD was shown to bind specifically to basic arginine- and lysine-rich rim patches on the core Sm domain (Hfq65) with a K_d of 2.9 μ M in low-salt buffer and 22 μ M in high-salt buffer. This intramolecular interaction acts as a nucleic acid mimic that competitively displaces non-specific RNA and dsRNA from the rim, recycling the chaperone and preventing non-specific DNA aggregation.

KEY FINDINGS (5)

The acidic CTD peptide specifically binds the Hfq core (Hfq65) with micromolar affinity.

protein_pair Hfq ↔ Hfq
evidence 2.9e-6 (quantitative)
context Fluorescence anisotropy of FITC-labeled CTD peptide binding to Hfq65 in low salt buffer
Kd 2.90e-06 M
residues Arg16, Arg17, Arg19, Lys47, Asp97, Glu99, Glu100, Glu102
confidence 0.95 · significant: True
source_span Page 4, Para 1

Ionic strength reduces binding affinity between the acidic CTD peptide and Hfq65 core.

protein_pair Hfq ↔ Hfq
evidence 2.2e-5 (quantitative)
context Fluorescence anisotropy of FITC-labeled CTD peptide binding to Hfq65 in higher salt buffer
Kd 2.20e-05 M
confidence 0.9 · significant: True
source_span Page 4, Para 1

E. coli Hfq65 binds a heterologous *Bacillus subtilis* CTD peptide with lower affinity.

protein_pair Hfq ↔ Hfq
evidence 8.7e-6 (quantitative)
context Fluorescence anisotropy titration of BsCTD-FITC peptide with *E. coli* Hfq65
Kd 8.70e-06 M
residues Lys3, Arg16, Arg17, Arg19
confidence 0.85 · significant: True
source_span Page 10, Para 2

Full-length Hfq102 exhibits lower binding affinity for dsRNA product than core Hfq65.

protein_pair Hfq ↔ Hfq
evidence 1.17e-7 (quantitative)
context Fluorescence anisotropy equilibrium binding to D16-FAM•R16 dsRNA complex at 30°C
Kd 1.17e-07 M
confidence 0.95 · significant: True
source_span Page 9, Para 1

Hfq-sCTD with shortened linker exhibits the weakest affinity for annealed dsRNA product.

protein_pair Hfq ↔ Hfq
evidence 1.81e-7 (quantitative)
context Fluorescence anisotropy equilibrium binding to D16-FAM•R16 dsRNA complex
Kd 1.81e-07 M
confidence 0.95 · significant: True
source_span Page 9, Para 1

CONTRADICTIONS AND NEGATIVE RESULTS (3)

FINDING	CONFLICTS WITH	REASONING
The mutated basic/polar CTD peptide (RSNKTN) was completely unable to bind Hfq65 core.	False	Acidic residues D97, E99, E100, and E102 at the CTD tip are strictly required for electrostatic binding to basic core rim residues.
Hfq distal face residue Lys31 is not contacted by the CTD.	False	Despite high solvent accessibility, Lys31 does not interact with the CTD, consistent with CTD selectively regulating the rim and proximal faces rather than distal face RNA binding.
The Hfq CTD does not directly bind DNA.	True	Previous reports suggested the CTD binds DNA, but rim mutant R16A abolished insoluble DNA aggregation with Hfq65, demonstrating dsDNA binds the core basic rim patch while the CTD modulates access.

METHODOLOGY

Methods

- Fluorescence anisotropy
- Stopped-flow fluorescence spectroscopy
- Electrophoretic mobility shift assay (EMSA)
- Rosetta FloppyTail de novo modeling
- Multiple sequence alignment and phylogenetic clustering
- Agarose gel electrophoresis DNA binding assay

Instruments

- Horiba Fluorolog-3
- Typhoon 9410

Controls

- Hfq65 truncation mutant lacking CTD
- Hfq-sCTD shortened linker mutant
- RSNKTN mutant CTD peptide lacking acidic residues
- Non-Hfq RNA annealing controls

Source organisms (NCBI taxon)

562, 287, 1423, 1280, 286

PDB accessions (5)

1HK9 1U1S 4NL2 3HSB 1KQ1

ENTITIES

Proteins

Hfq

Chemicals

FITC, TCEP, Cy3-maleimide, SYBR Gold

Equations $y = (K_a * x) / (1 + K_a * x)$; $y = \min Y + (\max Y - \min Y) / (1 + (x / IC_{50})^n)$; $C = 1 / ((V_{tail} - V_{core}) * NA)$; $ddG = -RT * \ln(Kd_{MUT} / Kd)$

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